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Roll No. ....

4th Sem / Civil

Subject : Surveying - II

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 When two tangents meet at a point, the point is known as _______

a) Apex b) Summit

c) Vertex d) None of the above

Q.2 What is the full form of G.I.S?

a) Geological Information System

b) Geographical Innovative System

c) Geological Innovative System

d) Geographical Information System

Q.3 When total station is sighted to the target, which of the operation acts first?

a) Rotation of optical axis.

b) Rotation of vertical axis

c) Rotation of horizontal axis

d) Rotation of line collimation

Q.4 Triangulation results consider to be more accurate while using _____ in DGPS.

a) Single frequency mode

b) Dual frequency mode

c) Random frequency mode

d) None of these

Q.5 A simple circular curve is designated by_______

a) Radius b) Length

c) Width d) None of the above

Q.6 Which among the following is a server based hardware platform of GIS?

a) Autodesk Revit b) STAAD Pro

c) Arc GIS d) Google-maps

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 A vertical curve having convexity upward is known as _______ (Tangential Curve / Summit curve)

Q.8 An instrument which is used to measure the area of the plan of any shape is known as _______ (Distomat / Planimeter)

Q.9 Vertical angle is measurement in the total station as zenith angle.(True/False)

Q.10 If we required the accuracy of positioning result within 10 to 15 meters then we use _____ (GPS / DGPS)

Q.11 There are control Panels in front and at the back of the total station. (True/False)

Q.12 DGPS can yield measurement good to a couple of meters in moving applications and even better in stationary situations.(True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Describe the necessity of providing the transition curves in highways and railways.

Q.14 Describe the use of various parts of a Total Station.

Q.15 Describe the procedure of "Prolonging a Line in Backward Direction" Using an electronic Digital Theodolite.

Q.16 Write a short note on GIS used in surveying.

Q.17 Write the application of remote sensing system in civil engineering surveying works.

Q.18 Write the uses of an EDM in civil engineering surveying works.

Q.19 Describe the various causes of errors in surveying works, while using a DGPS.

Q.20 Describe the use of various functional keys of a DGPS.

Q.21 Write a short note on "Applications of DGPS" in civil engineering surveying works.

Q.22 An instrument was set up at P and the angle of depression to a vane 4 m above the foot of the staff held at Q was 5o 36'. The horizontal distance between P and Q was known to be 3000 M. Determine the R.L . of the staff station Q, given that staff reading on a B.M. of elevation 436.050 was 2.865 m.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the procedure of temporary adjustment of an Electronic Digital Theodolite, with the help of diagram.

Q.24 Explain the procedure of measurement of vertical angles using a Total Station as a Surveying instrument.

Q.25 Two straights T_1V and Vt_2 of a road curve meet at an angle of 80^o. Find the radius of curve which will pass through a point P, 30 meter from the point of intersection (v), the angle T_1VP being 30^o. Also find the length of curve.